

# Changyu Zhao

UNDERGRADUATE STUDENT · COMPUTER SCIENCE

✉ thomasonzhao@outlook.com | 🌐 blog.thomasonzhao.cn | 📺 ThomasonZhao | 📄 changyu-zhao

## Education

### University of Wisconsin-Madison

Madison, WI, U.S.A

B.S. IN COMPUTER SCIENCE (HONORS) AND MATHEMATICS

Sep. 2021 - May 2025 (expected)

- Honors program in Computer Science with focus on research and academic excellence. Cumulative GPA: **3.96/4.0**.
- David Dewitt Undergraduate Scholarship (2024) for academic excellence and top undergraduate honor offered by the department.
- Thesis: *Advancing Dynamic Binary Analysis with Hardware Assisted Tracing*.
- Advisor: Prof. Patrick McDaniel

## Work Experience

### UW-Madison Cybersecurity Operation Center

Madison, WI, U.S.A

CYBERSECURITY STUDENT INTERN ANALYST

Feb. 2023 - Present

- Conducted security incident response and analysis for 400+ security incidents, including malware infections, phishing attack, unauthorized access attempts, and network intrusions, utilized Elastic search for log analysis, threat hunting and investigation, and recommending remediation actions.
- Managed monthly identity and access management (IAM) compliance attestation audits for 500+ employee accounts.

### Tencent Security Xuanwu Lab

Beijing, China

SECURITY RESEARCH INTERN

Jul. 2023 - Aug. 2023

- Engineered **LibIHT**, a cross platform dynamic binary tracing framework in C, C++, and Python to analyze control flow behavior of large programs with low overhead, leveraging Intel hardware tracing features such as Last Branch Record (LBR) and Branch Trace Store (BTS).
- Architected cross platform capabilities for Linux and Windows kernel driver to interface with Intel hardware features and provide full control on user-space tracing, implemented a unified set of API abstraction on kernel APIs for cross platform compatibility and avoiding duplicated code logic.
- Developed a user-space library to provide user-friendly high-level API access for customized user applications to interact with the tracing framework, implemented a plugin system for easy integration with debuggers like GDB and WinDbg to visualize and analyze the tracing data.

## Research Experience

### MadS&P - Security and Privacy Research Group at UW-Madison

Madison, WI, U.S.A

UNDERGRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. PATRICK MCDANIEL

Sep. 2024 - Present

- Conducting scientific measurements and evaluations of the performance advantages of **LibIHT** over existing dynamic binary tracing framework, focusing on the runtime overhead, tracing accuracy, and resilience to anti-tracing techniques.
- Developing a comprehensive evaluation methodology to assess the performance and effectiveness in real-world reverse engineering scenarios, including large-scale softwares and malware samples, to provide insights on practical implications of security research and industry applications.
- Contributing to a research paper to present the findings and insights from the performance evaluation, targeting top-tier security conferences.

### SEFCOM - Laboratory of Security Engineering for Future Computing at ASU

Tempe, AZ, U.S.A

UNDERGRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. YAN SHOSHITAISHVILI

May 2024 - Aug. 2024

- Assisted in the design and development of **Oxidizer**, a Rust-specific decompiler on top of *angr* C decompilation pipeline, aimed at producing Rust pseudo-code with recovered control flow structures, function prototypes, and common-used macros and struct data types from Rust binaries.
- Developed a Rust type scanner in Rust to collect type information from Rust standard libraries and popular crates, and integrated the type information into the decompilation process to recover Rust-specific data types and function signatures with pattern matching and type inference algorithms.
- Collected a comprehensive dataset of Rust binaries with source code to perform evaluations on the decompilation accuracy and effectiveness.

### SEFCOM - Laboratory of Security Engineering for Future Computing at ASU

Tempe, AZ, U.S.A

UNDERGRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. YAN SHOSHITAISHVILI

May 2023 - Jul. 2023

- Engineered **LibHopper**, an automated tool for discovering exploitation primitives of the internal state structures in user libraries.
- Designed analyzation pipeline consisting of debugger scripts to setup context for the internal state structures, and symbolic execution engine to evaluate security implications on memory corruption and control flow hijack, reducing manual analysis time and providing structured assessments.
- Concluded a comprehensive dataset of potential exploitation primitives and possible mitigation strategies in popular user libraries, including *openssl*, *openssh*, *libcurl*, and *libpng*, etc., contributed to the development of automated exploit generation tools and vulnerability scanners.

# Extracurricular Activity

## r3kapiG CTF Team

TEAM MEMBER

International

Sep. 2021 - Present

- Competed in over 40 high-quality 48-hour Capture The Flag (CTF) competitions. Expert in **binary exploitation** and **reverse engineering**.
- Entered DEFCON CTF world final competitions from 2022 to 2024, ranked 12th, 7th, and 4th respectively.

## Cybersecurity UW Club

Madison, WI, U.S.A

CORE CTF TEAM MEMBER, LECTURER

Sep. 2021 - Present

- Led the Cybersecurity UW CTF team to top 2% in a national cybersecurity competition, providing strategic guidance and technical expertise.
- Delivered over 10 workshop lectures on **binary exploitation** and **reverse engineering**, contributing to the skill development of 30+ club members.
- Deepened technical skills while contributing to the broader cybersecurity community through leadership, mentoring, and knowledge sharing.

## pwn.college

Online

INDEPENDENT LEARNER, VOLUNTEER PEER MENTOR

Dec. 2021 - Present

- Attained **Yellow Belt** and **Blue Belt** certifications by completing all challenges in the pwn.college, currently ranked **43 out of 2500+** participants.
- Developed comprehensive skills in software and system security, including reverse engineering, software exploitation, and kernel security.
- Mentored and guided students, particularly from underrepresented groups, by helping them build foundational skills in cybersecurity.
- Expanded access to cybersecurity education by actively participating in the learning community, answering questions, and offering advice.

# Honors & Awards

|         |                                                                              |                      |
|---------|------------------------------------------------------------------------------|----------------------|
| 2021-24 | <b>Dean's List</b> , UW-Madison College of Letters & Science                 | Madison, WI, U.S.A   |
| 2024    | <b>David Dewitt Undergraduate Scholarship</b> , UW-Madison Computer Sciences | Madison, WI, U.S.A   |
| 2024    | <b>Finalist</b> , DEFCON 32nd CTF World Final                                | Las Vegas, NV, U.S.A |
| 2023    | <b>Finalist</b> , CSAW'23 CTF Final (US-Canada region)                       | New York, NY, U.S.A  |
| 2023    | <b>Finalist</b> , DEFCON 31st CTF World Final                                | Las Vegas, NV, U.S.A |
| 2022    | <b>Finalist</b> , DEFCON 30th CTF World Final                                | Las Vegas, NV, U.S.A |
| 2022    | <b>49/3658, Top 2%</b> , Cyber Skyline National Cyber League Team Game       | Madison, WI, U.S.A   |

# Presentation

## BlackHat USA

Las Vegas, NV, U.S.A

ARSENAL PRESENTER FOR "LIBIHT: A CROSS-PLATFORM LIBRARY FOR ACCESSING INTEL HARDWARE TRACE FEATURES"

Aug. 2024

- Presented **LibIHT** to industry professionals, highlighting its advanced tracing capabilities in dynamic control flow analysis on large programs.
- Discussed use cases for LibIHT in vulnerability research, showcasing its potential to improve analysis capabilities for reverse engineering tasks.

# Skills

|                            |                                                                                 |
|----------------------------|---------------------------------------------------------------------------------|
| <b>Programming</b>         | Advanced: C/C++, Python, Java, Assembly; Intermediate: Node.js, LaTeX           |
| <b>Reverse Engineering</b> | Advanced: IDA Pro, Ghidra, GDB, Windbg; Intermediate: Angr, QEMU, Frida, Joern  |
| <b>Network Security</b>    | Advanced: Burp Suite, Wireshark; Intermediate: Metasploit Framework, Kali Linux |
| <b>Languages</b>           | Chinese (Native), English (Fluent)                                              |